



Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Course No : CSE 2214
Course Title : Assembly Language Programming Sessional
Assignment No : 06

Date of Submission : Thursday, 1st October, 2026

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Submitted By-

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Section : A

Write an assembly language program that displays a sequence formed by interleaving the last two digits of your studentID with the last two letters of your name(digit-letter-digit-letter-...), one by one, on consecutive dot matrix displays. After the sequence has been shown, the program should continuously scroll the pattern from right to left until the entire sequence reaches the leftmost position, then reverse the direction and scroll it from left to right. Repeat the process continuously with suitable delays.

For example, If the Student's name is Ahmed Abdullah, and his ID is 00724105101027, the sequence to be displayed should be 2A7H

```
;00724105101024  
MOV BX,-1
```

L1:

```
LEA SI,ARR  
MOV DX,PORT1  
MOV CX,20
```

L2:

```
MOV AL,[SI]  
OUT DX,AL  
INC SI  
INC DX  
LOOP L2
```

```
ADD PORT1,BX  
CMP PORT1,2000H  
JL MOV_RIGHT2  
CMP PORT1,2014H  
JG MOV_LEFT2
```

```
CMP BX,1  
JE MOV_RIGHT1  
CMP BX,-1  
JE MOV_LEFT1
```

MOV_LEFT1:

```
MOV DX, PORT1  
CALL DELAY  
ADD DX, 20  
MOV AL, 0000000B
```

```
OUT DX,AL
CALL DELAY
JMP L1
```

```
MOV_RIGHT1:
CALL DELAY
MOV DX,PORT1
DEC DX
MOV AL,00000000B
OUT DX,AL
CALL DELAY
JMP L1
```

```
MOV_LEFT2:
MOV PORT1,2013H
MOV BX,-1
MOV DX,2027H
MOV AL,00000000B
OUT DX,AL
CALL DELAY
JMP L1
```

```
MOV_RIGHT2:
MOV PORT1,2001H
MOV BX,1
MOV DX,2000H
MOV AL,00000000B
OUT DX,AL
CALL DELAY
JMP L1
```

```
ARR DB
1111001B,1001001B,1001001B,1001001B,1001111B,1111111B,0001001B,0001001B,0000001B,
0000001B,0001111B,0001000B,0001000B,0001000B,1111111B,1000001B,1000001B,1111111B,
1000001B,1000001B
T DB 0
PORT1 DW 2014H
PORT2 DW 2000H
```

```
DELAY PROC
MOV T,20
L5:
DEC T
CMP T,0
JG L5
RET
DELAY ENDP
```